

Data Science & Machine Learning 2 – Preliminary Submission Report

Machine Learning Portfolio

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1. Overview

This document is my preliminary submission for Data Science & Machine Learning 2. It presents the current state of my machine learning portfolio and outlines the work I have completed so far.

The portfolio includes implementations of several key machine learning topics covered in the module. Rather than focusing on only one model, I developed a set of notebooks that demonstrate different approaches, including supervised learning, unsupervised learning, neural networks, reinforcement learning, and deep learning.

For most of the notebooks, I used a dataset about the impact of AI on student performance. This dataset allows classification and clustering tasks using academic and AI-related features. For reinforcement learning, I used the FrozenLake environment from Gymnasium, since reinforcement learning requires an environment-based setup rather than a standard labelled dataset.

This submission includes the completed notebooks, supporting log files, a README file, the dataset used, and an outline of the remaining work planned before the final submission.

2. Work Completed So Far

At this stage, I have completed the following parts of the portfolio.

2.1 K-Nearest Neighbour (KNN)

I created a notebook called KNN.ipynb that implements the k-Nearest Neighbour algorithm as a classification model.

This work includes:

- loading and inspecting the AI and student performance dataset
- handling missing values in categorical columns
- selecting performance_category as the target variable
- encoding categorical variables using LabelEncoder
- splitting the data into training and testing sets
- scaling the input features using StandardScaler
- training and evaluating a KNN classifier
- comparing different values of k
- selecting the best-performing value of k

The notebook showed that KNN could classify performance levels reasonably well, and it also demonstrated the importance of scaling and parameter tuning.

2.2 K-Means Clustering

I created KMeans.ipynb to implement an unsupervised learning example using k-Means clustering.

This work includes:

- loading and preparing the dataset
- removing fields that should not be used for clustering
- encoding categorical variables
- scaling the features
- using the elbow method to inspect suitable values of k

- training a k-Means clustering model
- assigning cluster labels to the data
- comparing clusters with the actual performance categories
- visualising clusters using PCA

This notebook demonstrated how clustering differs from classification. The resulting clusters did not strongly match the labelled performance groups, but they showed useful behavioural grouping patterns, especially around AI usage.

2.3 Support Vector Machine (SVM)

I created SVM.ipynb to implement Support Vector Machine classification.

This work includes:

- preparing the same AI and student performance dataset for classification
- encoding categorical variables
- scaling the features
- training an initial SVM model
- evaluating results using accuracy, classification report, and confusion matrix
- comparing different kernels, including linear, RBF, polynomial, and sigmoid
- selecting the best-performing kernel

This notebook produced one of the strongest supervised learning results in the portfolio. It also demonstrated how kernel choice can affect model performance.

2.4 Neural Networks

I created NeuralNetworks.ipynb to implement an Artificial Neural Network using MLPClassifier.

This work includes:

- preparing the dataset for multi-class classification

- encoding categorical variables
- scaling the features
- training an initial neural network model
- evaluating model performance
- comparing different hidden layer settings
- selecting the best-performing network structure
- analysing the effect of architecture size on performance

This notebook showed that neural networks can perform well on the dataset, while also showing that larger architectures do not always lead to better accuracy.

2.5 Reinforcement Learning

I created ReinforcementLearning.ipynb to implement a reinforcement learning example using Q-learning and the FrozenLake environment from Gymnasium.

This work includes:

- creating the FrozenLake environment
- identifying the available states and actions
- creating and initialising a Q-table
- setting Q-learning parameters such as learning rate, discount factor, exploration rate, and number of episodes
- training the agent over many episodes
- updating the Q-table using the Q-learning rule
- tracking reward progress over time
- testing the trained agent after learning
- visualising the final Q-table

This notebook was particularly useful because it is clearly different from the other portfolio items. It showed how reinforcement learning works through interaction, rewards, and repeated experience rather than labelled examples.

2.6 Deep Learning

I created `DeepLearning_Transformers.ipynb` to implement a deep learning model using TensorFlow/Keras.

This work includes:

- preparing the dataset for multi-class classification
- encoding categorical variables
- scaling the input features
- converting target labels into one-hot encoded format
- building and training a dense deep learning model
- evaluating the first deep learning model
- comparing several alternative deep learning architectures
- analysing the impact of model depth and width
- plotting loss and accuracy curves

Although the notebook title refers to deep learning and transformers, the implementation focuses on deep learning rather than transformers. This is because the dataset is tabular rather than text- or sequence-based, so a dense neural architecture is more suitable than a transformer-based model for this task.

3. Current Portfolio Structure

The portfolio is currently organised into the following main sections:

- `data/` – contains the dataset used in the portfolio
- `notebooks/` – contains all machine learning notebooks
- `logs/` – contains a log file for each notebook or algorithm
- `README.md` – contains an overview of the portfolio
- `requirements.txt` – lists the Python dependencies used in the project
- `preliminary_report.pdf` – contains this preliminary report

The current notebooks included are:

- `KNN.ipynb`
- `KMeans.ipynb`
- `SVM.ipynb`
- `NeuralNetworks.ipynb`

- ReinforcementLearning.ipynb
- DeepLearning_Transformers.ipynb

This structure keeps the work organised and clearly separates the data, notebooks, logs, and supporting documentation.

4. Progress Logging

To meet the assessment requirements, I created separate log files for each main notebook or algorithm:

- KNN_log.md
- KMeans_log.md
- SVM_log.md
- NeuralNetworks_log.md
- ReinforcementLearning_log.md
- DeepLearning_Transformers_log.md

These logs record:

- the work completed
- changes or alterations made
- the practical impact of those changes
- analysis and reflection on the development process

The logs are an important part of the portfolio because they show not only the final results, but also how the work developed over time and how problems were identified and solved.

5. Technologies and Tools Used

The portfolio uses the following main tools and libraries:

- Python
- Jupyter Notebooks
- pandas

- numpy
- matplotlib
- seaborn
- scikit-learn
- TensorFlow / Keras
- Gymnasium

I chose these tools because they are widely used for machine learning and data science, and they support different types of tasks such as classification, clustering, neural networks, reinforcement learning, and deep learning.

6. Reflection on Current Progress

At this stage, the portfolio already covers a broad range of machine learning topics. It includes examples of:

- supervised learning
- unsupervised learning
- neural networks
- reinforcement learning
- deep learning

This gives the portfolio more variety and better alignment with the module brief than focusing on only one problem or one algorithm.

One important strength of the current portfolio is that it does not only show final working results, but also shows development and improvement. For example:

- the KNN notebook includes tuning of the value of k
- the SVM notebook compares different kernel choices
- the neural network notebook compares different hidden layer settings
- the reinforcement learning notebook includes a first unsuccessful attempt followed by improved parameter tuning
- the deep learning notebook compares multiple deep learning architectures and analyses performance differences

This process demonstrates practical learning and problem solving, which is an important part of the portfolio.

7. Planned Work for Final Submission

Before the final submission, I plan to complete the following tasks:

- review and clean all notebooks so they run clearly from top to bottom
- improve markdown explanations and notebook presentation where needed
- refine and polish all log files
- improve the README file so it gives a stronger overview of the whole portfolio
- update requirements.txt so it fully reflects the libraries used
- review the final results and make sure they are described consistently across notebooks, logs, and report
- prepare the final portfolio submission in a clean and professional structure
- finalise and practise the short presentation

The goal for the final submission is to present a complete and professional portfolio that clearly demonstrates the range of machine learning topics covered and the practical work completed.

8. Conclusion

This preliminary submission shows that the portfolio has already developed into a broad machine learning collection rather than a single-model project. The completed work now includes classification, clustering, neural networks, reinforcement learning, and deep learning.

So far, the portfolio demonstrates practical skills in:

- data preparation
- preprocessing
- classification
- clustering
- model comparison
- neural networks
- reinforcement learning
- deep learning

- evaluation and reflection

The remaining work is mainly to refine, polish, and present the portfolio more clearly for the final submission. The main technical implementations are already in place, and the next step is to strengthen the documentation, consistency, and presentation quality of the overall submission.